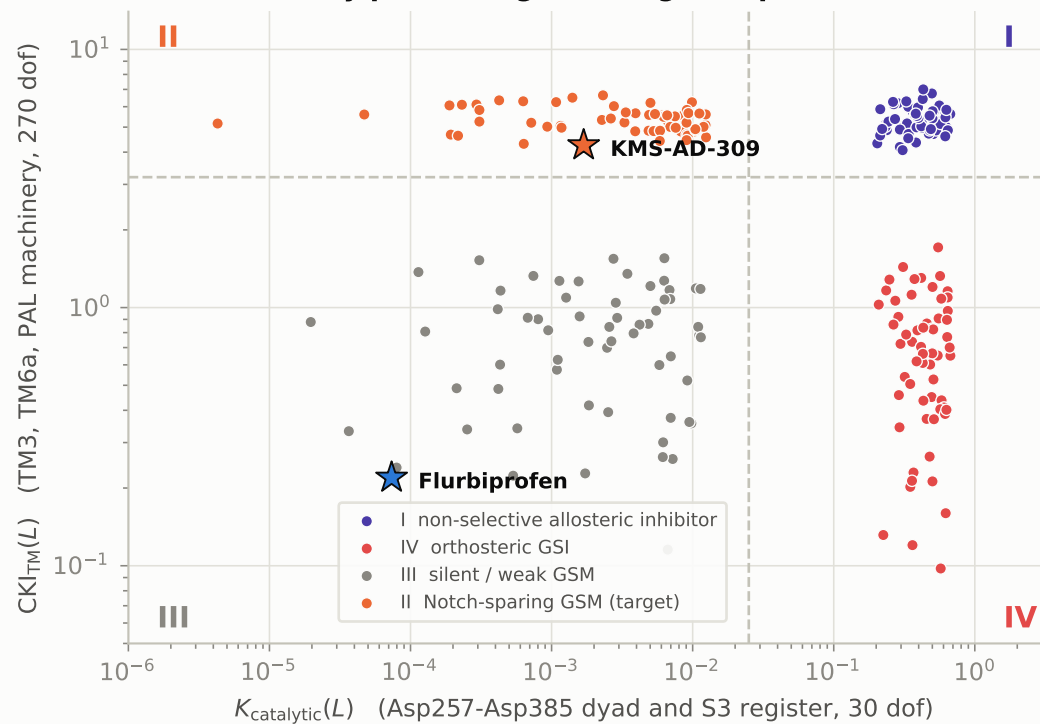
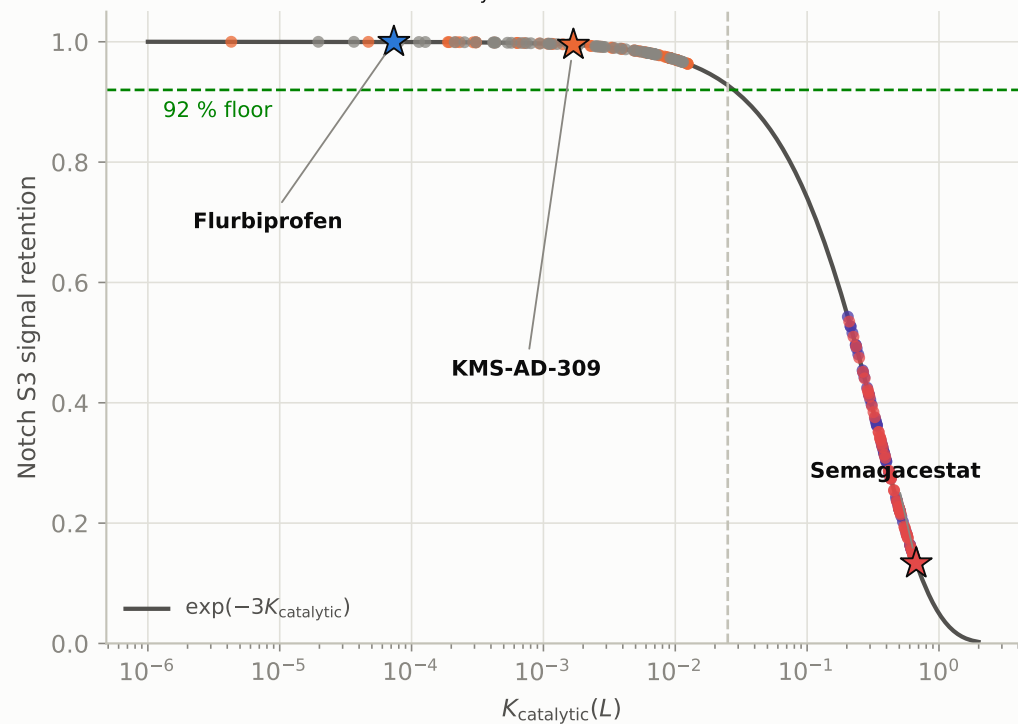
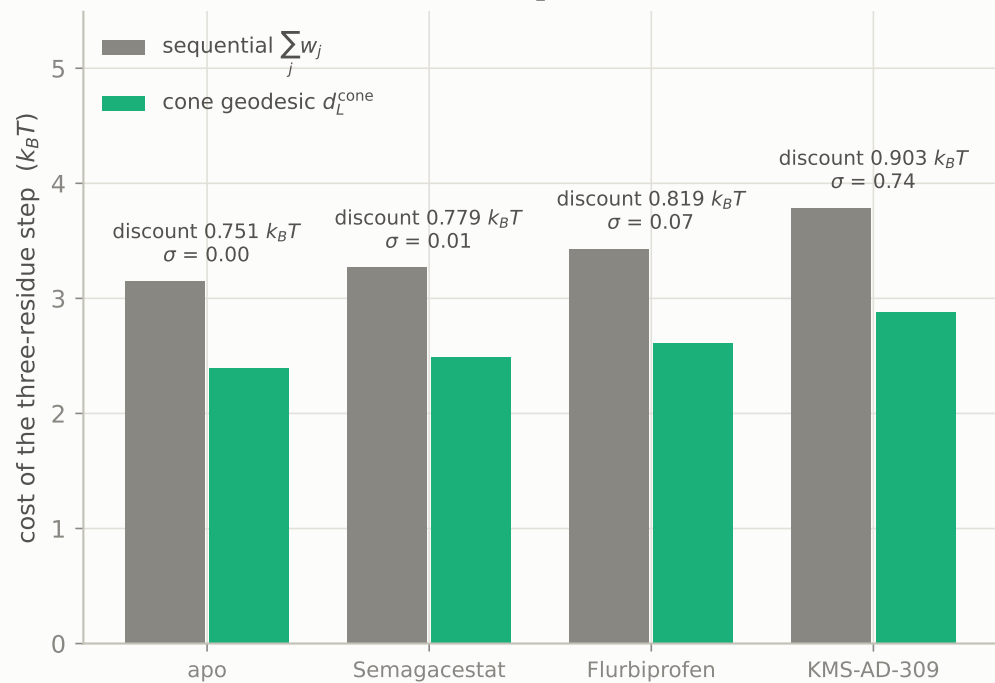


A Notch-safety phase diagram; target = quadrant II**B Notch constraint: $K_{\text{catalytic}} < 0.025 \Rightarrow \text{retention} > 92\%$** **C Cone geodesic on $\{0, 1\}^3$: $d_L^{\text{cone}} = 2.284457 w < 3w$** **D Pathogenic-ratio law; untreated R_0 dashed**